

JASON FORCE

Perception • Robotics • Navigation Systems

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Professional Summary

Perception, robotics, and navigation systems engineer with expertise spanning computer vision, computer vision target tracking, visual inertial navigation systems (VINS), SLAM, controls, and heterogeneous computing architectures. Experienced developing and integrating sensing, software, firmware, and electromechanical systems for UAV, robotics, and advanced autonomy applications. Background includes motion estimation, perception system development, precision tracking, and multidisciplinary integration for real-world robotic systems.

Core Technical Expertise

Tracking & Perception: Computer vision target tracking, motion estimation, visual tracking

Computer Vision & AI: PyTorch, CNNs, OpenCV, Open3D, Scikit-learn, machine learning

Robotics & Navigation: ROS 2, VINS-Fusion, PX4, MAVROS, MAVLink, Gazebo, GTSAM with ISAM, SLAM, autonomous navigation

Controls & Estimation: Kalman filtering, factor graphs, state estimation, sensor fusion, feedback control

Software & Firmware: Python, C++, firmware, Embedded Linux, RTOS, ARM architectures, asynchronous systems, real-time control

Optics, Motion & Mechatronics: Optomechanical design, precision motion systems, frameless motors, motor drive systems, precision bearings, pointing systems

Hardware & Processing: CPU/GPU/TPU/DSP/FPGA architectures, microprocessors, AI coprocessors, Intel RealSense, Ambarella, Hailo

Engineering & Development Tools: Git, VSCode, PyCharm, Jupyter Notebooks, Google Colab, Amazon SageMaker, Claude, Siemens NX

Technical Communications & Planning: Technical documentation, system architecture communication, Confluence, Jira, EaseRequirements, requirements traceability, multidisciplinary technical coordination

Professional Experience

Software Lead

Overview Energy | Sep 2024 – Present

- Serve as technical lead supporting spacecraft software, subsystem design, and cross-domain technical development for space-based energy systems.
- Participate in system-level architecture trade studies and interface analysis for spacecraft bus-to-payload communications, including CCSDS and MaxPackets approaches.
- Perform optomechanical and mechanical design using Siemens NX, supporting precision motion system development including frameless motors, precision bearings, and pointing-related mechanisms.

Mobile Robotics Perception Engineer

Force Consulting Services LLC | Dec 2023 – Sept 2024

- Developed and evaluated visual inertial navigation system (VINS) approaches for UAV localization and navigation in varied operating environments.
- Developed geo-referenced imagery navigation and VINS-based SLAM systems.
- Integrated ROS 2, PX4, MAVROS, MAVLink, and Gazebo-based simulation and autonomy workflows.
- Evaluated perception, target tracking, and autonomy approaches for robotics and navigation applications.

MLOps Tool Developer

Amazon Fulfillment Technologies and Robotics (Contractor) | May 2023 – Dec 2023

- Developed dataset conversion tools supporting Ambarella vision processor integration.
- Converted custom datasets for YOLO training workflows in Amazon SageMaker.
- Integrated conversion and training pipelines into Jupyter notebook environments.
- Supported machine vision model training and deployment workflows.

Machine Vision Engineer

Force Consulting Services LLC for United States Postal Service | Nov 2020 – May 2023

- Developed AI and computer vision algorithms for logistics and automation applications.
- Implemented RNN event processing for infrared range camera systems.
- Evaluated ORB-SLAM3 for facility navigation and robotic guidance.
- Developed machine vision systems for container fullness measurement and automated mail sorting.
- Built perception pipelines using Python, OpenCV, Open3D, asyncio, and related tooling.

Director of Imaging Systems

Applied Signals Intelligence Inc. | May 2019 – Nov 2020

- Developed UAV vision-based pose estimation software using ground fiducials.
- Created software interfaces for camera systems and tracking hardware.
- Developed image processing applications on Raspberry Pi platforms.
- Supported passive optical tracking and navigation applications.

President & Principal Engineer

Force Labs Inc. | Feb 2012 – Aug 2019

- Led multidisciplinary consulting and prototyping efforts for government and industry customers spanning firmware, sensing, controls, and electromechanical systems.
- Architected and developed hardware and software systems including PCB-based electronics, firmware, Linux-based systems, sensor integration, and robotics-related subsystems.
- Developed stereo vision, sensing, controls, and electromechanical solutions for advanced technical applications.
- Managed technical proposals, engineering coordination, and system-level architecture decisions.

Founder

Iron Goat Technology Inc. | Feb 2013 – May 2016

- Developed biomass-powered mobile robotics systems.
- Developed stereo vision guidance systems using ROS, OpenCV, and embedded computing platforms.
- Designed brushless motor drive systems and supporting electronics.
- Led system-level robotics architecture spanning communications, sensing, motion systems, and perception.

Education

M.S., Electrical Engineering — George Mason University, Fairfax, VA (2017)

B.S., Electrical Engineering — George Mason University, Fairfax, VA (1995)

Professional Development

Selected coursework in computer vision, AI, robotics, and related technologies.